

UNIVERSITY OF ASIA PACIFIC



Department of Computer Science & Engineering

Course Code: CSE 404

Course Title: Artificial Intelligence and Expert Systems Lab

Topic : Implement the Multivariable Linear Regression using Open Source Dataset with and without Scikit Learn.

Submitted By:

Iffat Rahman Etu(21201168)

Most.Nasrin Akter Nipa(21201177)

Md. Fahad Prodhan(21201151)

Problem Title: Implement the Multivariable Linear Regression using Open Source Dataset with and without Scikit Learn.

Problem Description: In this problem, we aim to implement multivariable linear regression using Scikit Learn and without Scikit Learn using any external libraries, leveraging an open-source dataset. Linear regression is a fundamental technique used for modeling the relationship between multiple independent variables (features) and a single dependent variable (target).

Tools and Languages Used:

➤ **Google Colab**



➤ **Python Language**



Multivariate Linear Regression:

Multivariate linear regression, also known as multivariable linear regression or multiple linear regression, is a statistical method used to analyze the relationship between multiple independent variables (also known as predictors or features) and a dependent variable (also known as the response or target variable).

In simple linear regression, we examine the relationship between a single independent variable and a dependent variable. But here we analyze the effect of two or more independent variables on a dependent variable.

In multivariate linear regression, the relationship between the independent variables and the dependent variable is represented by the following equation :

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon$$

Where:

- ✓ y is the dependent variable (target).
- ✓ $x_1, x_2, \dots, x_n$ are the independent variables (features)..
- ✓ β_0 is the intercept term (the value of y when all independent variables are zero).
- ✓ $\beta_1 + \beta_2 + \dots + \beta_n$ are the coefficients (slope parameters) representing the relationship between each independent variable and the dependent variable.
- ✓ ϵ is the error term, representing the difference between the predicted and actual values of the dependent variable.

- ✓ The goal of multivariate linear regression is to estimate the coefficients $\beta_1, \beta_2, \dots, \beta_n$ between the predicted and actual values of the dependent variable that best fit the observed data, minimizing the sum of squared variable.

Multivariate Linear Regression Cost Function:

The cost function for multivariate linear regression, often referred to as the mean squared error (MSE) or the residual sum of squares (RSS), measures the difference between the predicted values and the actual values of the dependent variable. For a dataset with m samples and n features, the cost function $J(\beta)$ for multivariate linear regression is given by:

$$J(\beta) = \frac{1}{2m} \sum_{i=1}^m (h_{\beta}(x^{(i)}) - y^{(i)})^2$$

Where:

- β is the vector of coefficients, including the intercept term
- β_0 and the coefficients for each feature $\beta_1, \beta_2, \dots, \beta_n$
- m is the number of samples in the dataset.
- $x_{(i)}$ represents the feature vector of the i^{th} sample.
- $y^{(i)}$ is the actual value of the dependent variable for the i^{th} sample.

Gradient Descent Algorithm:

The gradient descent algorithm is an optimization technique used to minimize a function by iteratively moving in the direction of the steepest descent (negative gradient) of the function. It's commonly used in machine learning for training models, particularly in cases where the cost function is convex or can be approximated as convex.

Compute the gradient of the mean squared error (MSE) cost function with respect to each parameter θ_j using the formula:

$$\frac{\partial}{\partial \theta_j} J(\theta) = \frac{1}{m} \sum_{i=1}^m (h\theta(x^{(i)}) - y^{(i)}) \cdot x_j^{(i)}$$

Where:

- m is the number of samples in the dataset.
- $(h\theta(x^{(i)}))$ is the predicted value for sample $(x^{(i)})$ using the current parameters θ .
- $y^{(i)}$ is the actual value of the dependent variable for sample $x^{(i)}$.
- $x_j^{(i)}$ is the value of the j^{th} feature for sample $x^{(i)}$.

Update each parameter θ_j using the gradient and the learning rate α according to the following equation:

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

Diamond Price Prediction Dataset:

Link : [Diamond price dataset link](#)

We have downloaded the dataset from Kaggle.

In our dataset, we have 53940 rows in the dataset and 11 columns.

Target Variable : Price

- ✓ Unnamed
- ✓ carat
- ✓ cut
- ✓ color
- ✓ clarity
- ✓ depth
- ✓ table
- ✓ price
- ✓ x
- ✓ y
- ✓ z

This classic dataset contains the prices and other attributes of almost 54,000 diamonds. There are 10 attributes included in the dataset including the target ie. price.

carat (0.2-5.01): The carat is the diamond's physical weight measured in metric carats. One carat equals 0.20 gram and is subdivided into 100 points.

cut (Fair, Good, Very Good, Premium, Ideal): The quality of the cut. The more precise the diamond is cut, the more captivating the diamond is to the eye thus of high grade.

color (from J (worst) to D (best)): The colour of gem-quality diamonds occurs in many hues. In the range from colourless to light yellow or light brown. Colourless diamonds are the rarest. Other natural colours (blue, red, pink for example) are known as "fancy," and their colour grading is different than from white colorless diamonds.

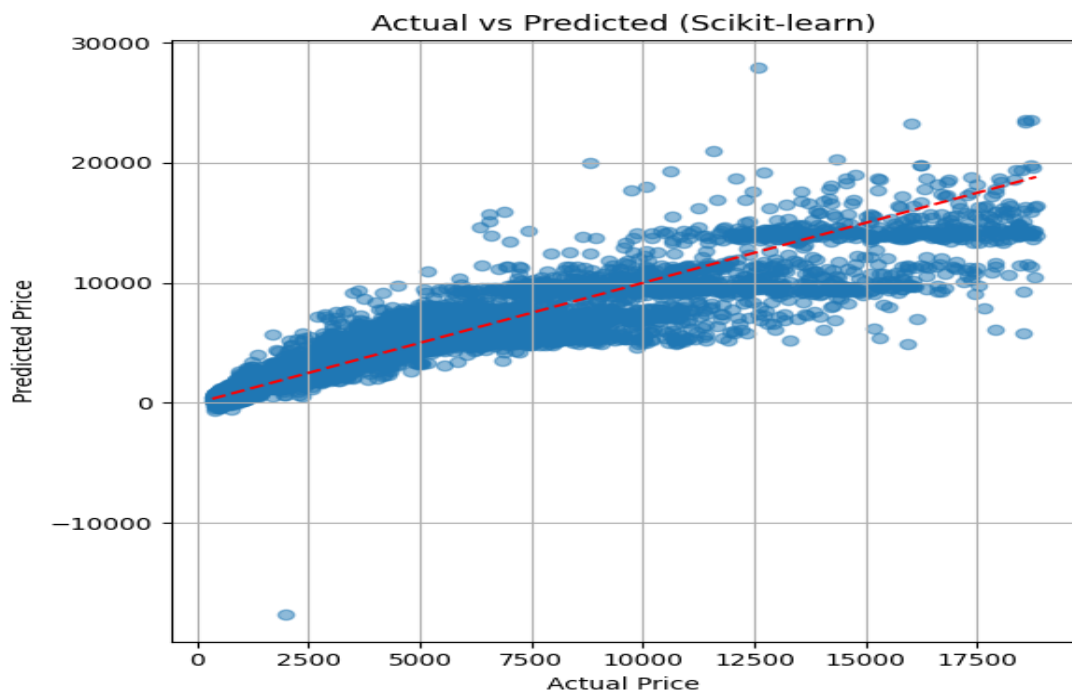
clarity (I1 (worst), SI2, SI1, VS2, VS1, VVS2, VVS1, IF (best)): Diamonds can have internal characteristics known as inclusions or external characteristics known as blemishes. Diamonds without inclusions or blemishes are rare; however, most characteristics can only be seen with magnification.

depth (43-79): It is the total depth percentage which equals to $z / \text{mean}(x, y) = 2 * z / (x + y)$. The depth of the diamond is its height (in millimetres) measured from the culet (bottom tip) to the table (flat, top surface) as referred in the labelled diagram above.

table (43-95): It is the width of the top of the diamond relative to widest point. It gives diamond stunning fire and brilliance by reflecting lights to all directions which when seen by an observer, seems lustrous.

price (\$\$326 - \$18826): It is the price of the diamond in US dollars. It is our very target column in the dataset. **x** (0 - 10.74): Length of the diamond (in mm) **y** (0 - 58.9): Width of the diamond (in mm) **z** (0 - 31.8): Depth of the diamond (in mm)

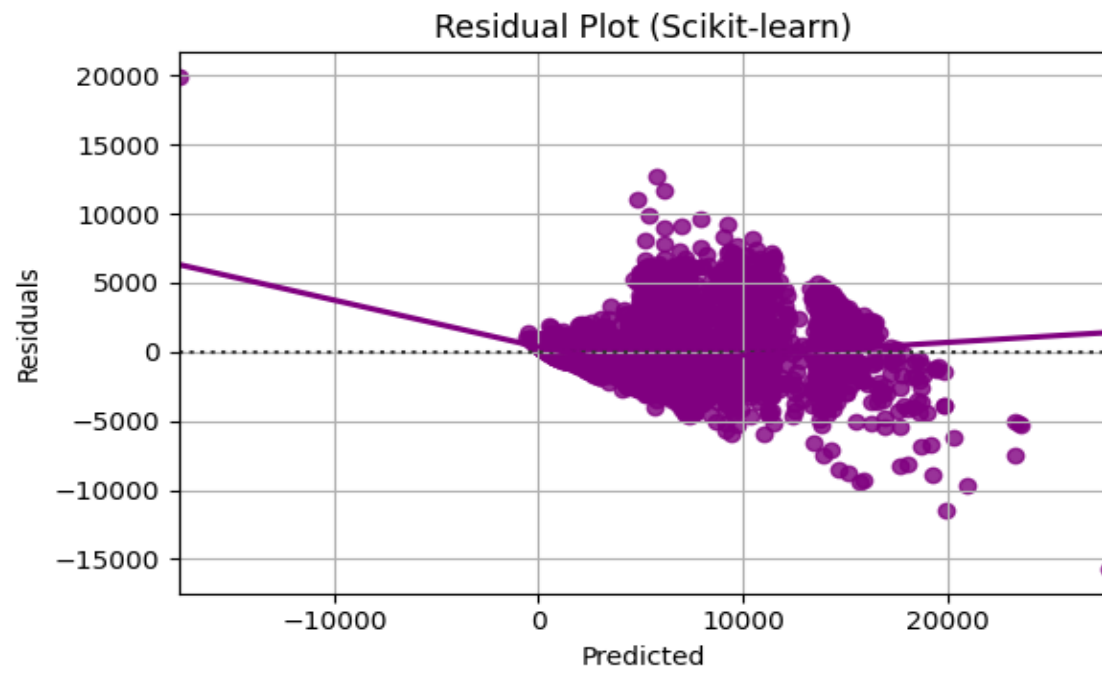
Diamond Price Prediction using SK-Learn:



Residual plot:

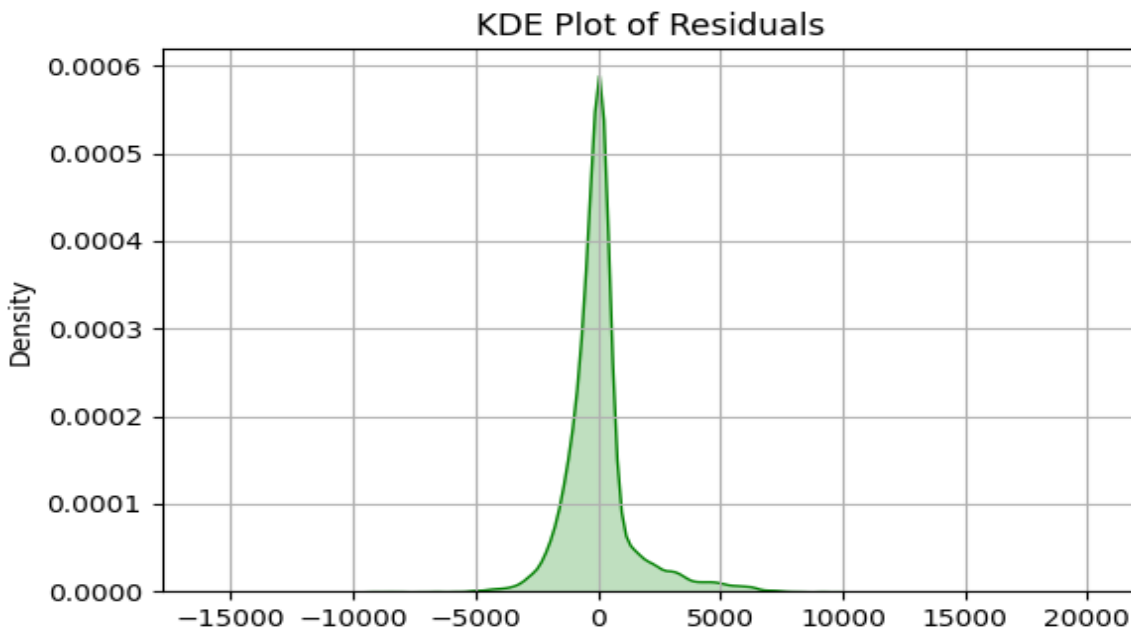
The residual plot shows the difference between the actual and predicted values (residuals) on the y-axis, and the predicted values on the x-axis. In the Scikit-learn model, the residuals are randomly scattered around zero, which indicates that the model fits the data well and there is no obvious

pattern or bias.



KDE plot :

The KDE (Kernel Density Estimation) plot of residuals in the Scikit-learn model displays a smooth bell-shaped curve centered around zero, indicating that the residuals are approximately normally distributed. This supports the assumption that the errors are random and independent, which is ideal for linear regression.

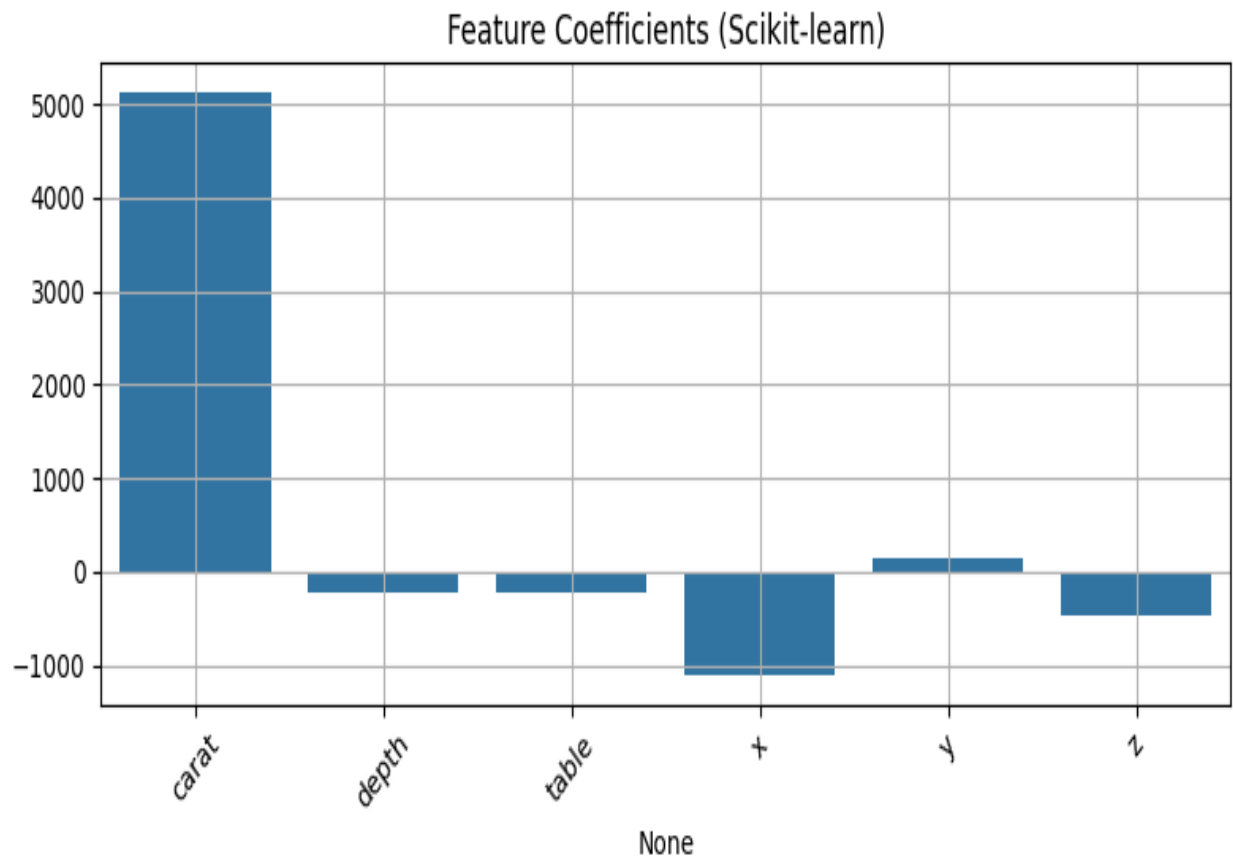


Coefficient Bar Plot:

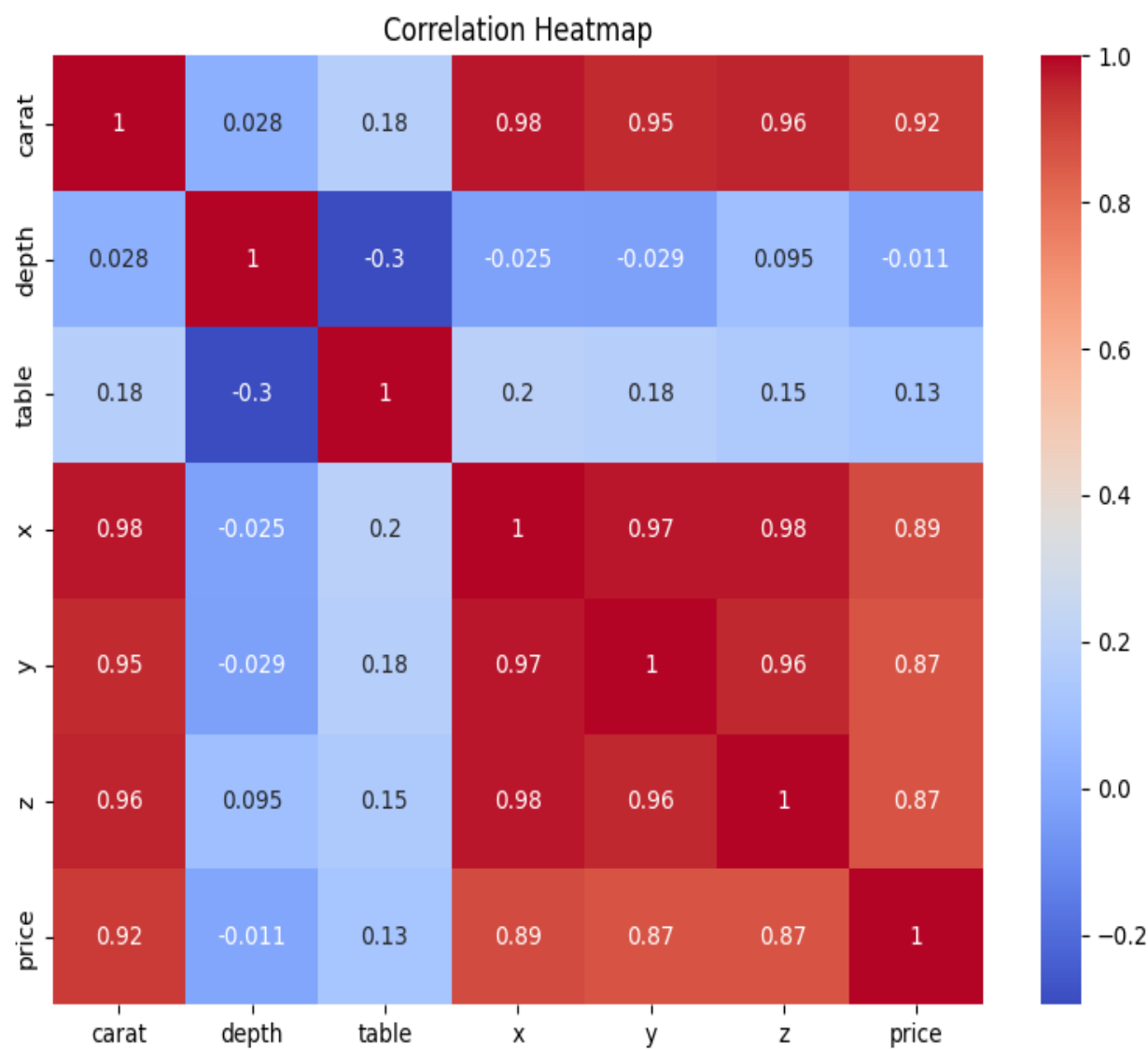
The coefficient bar plot displays the impact of each independent feature on the target variable (price). In the Scikit-learn model, each bar represents a feature's learned coefficient.

- Positive coefficients indicate a direct relationship with price (as the feature increases, price increases).
- Negative coefficients indicate an inverse relationship with price (as the feature increases, price decreases).

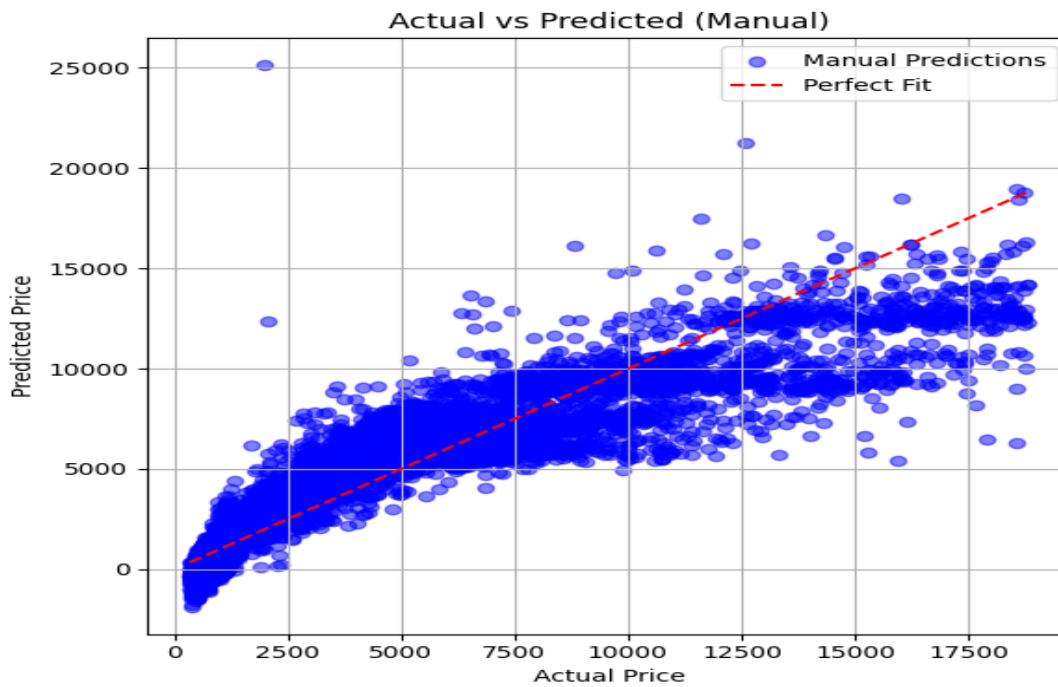
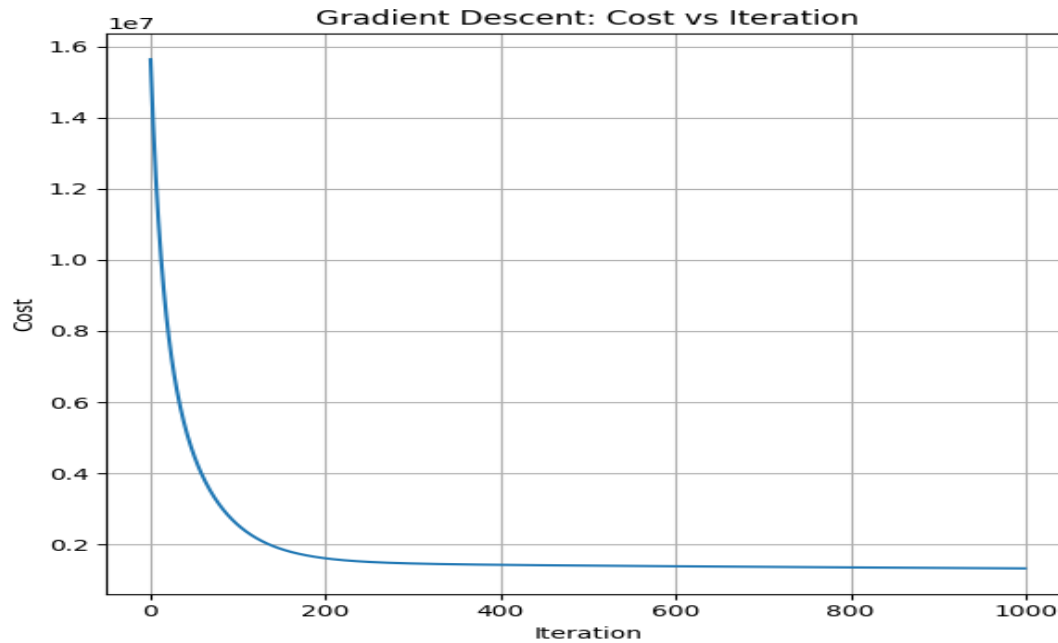
The plot provides a clear visual comparison of how strongly and in which direction each feature influences the model's predictions.



Heatmap :

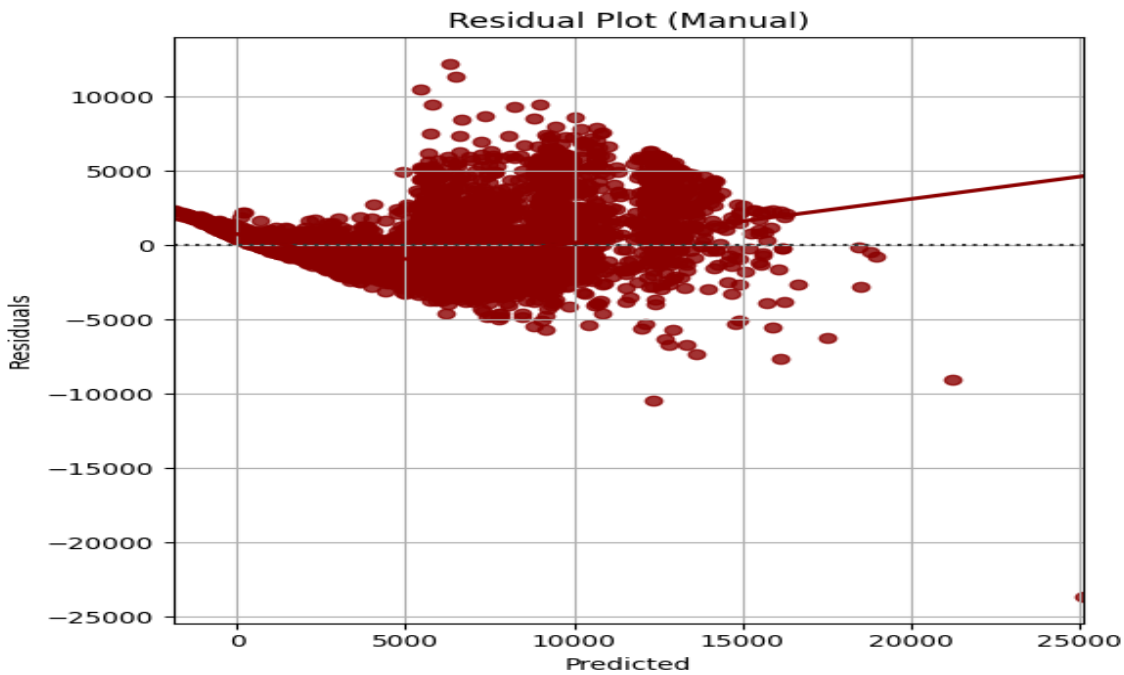


Diamond Price Prediction without Scikit Learn:



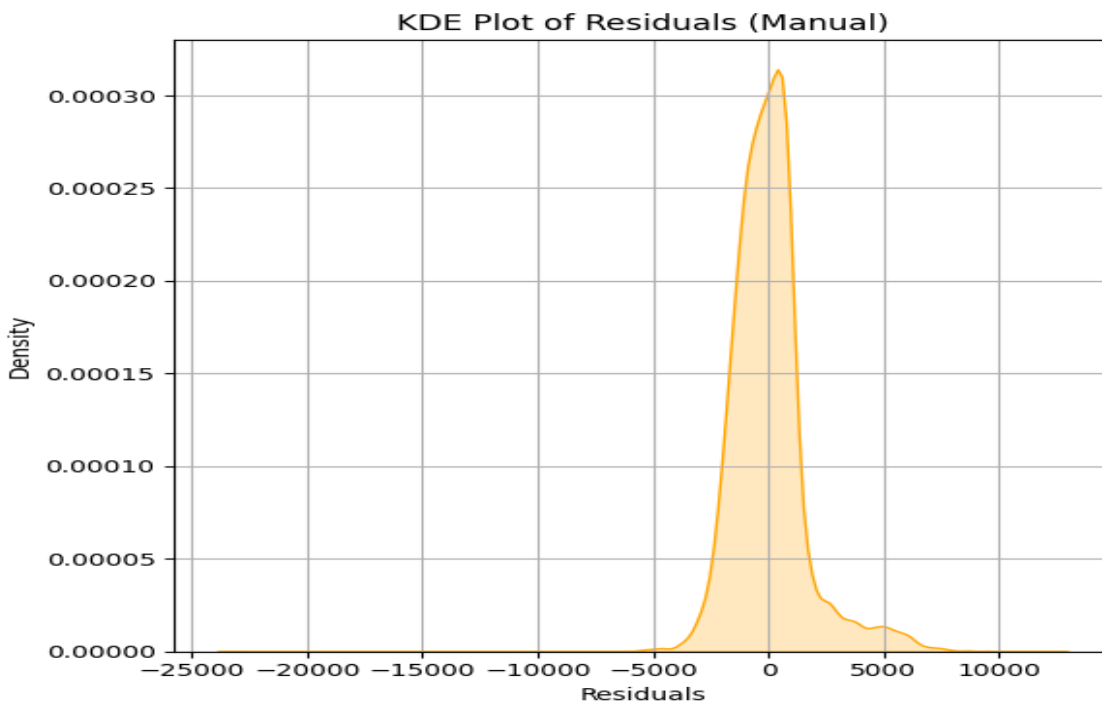
Residual Plot :

In the manually implemented model (without Scikit-learn), the residual plot also shows a similar pattern, but there may be slightly more spread or uneven clustering of residuals compared to the Scikit-learn model. This suggests the model is generally accurate but not as precise as the Scikit-learn version.

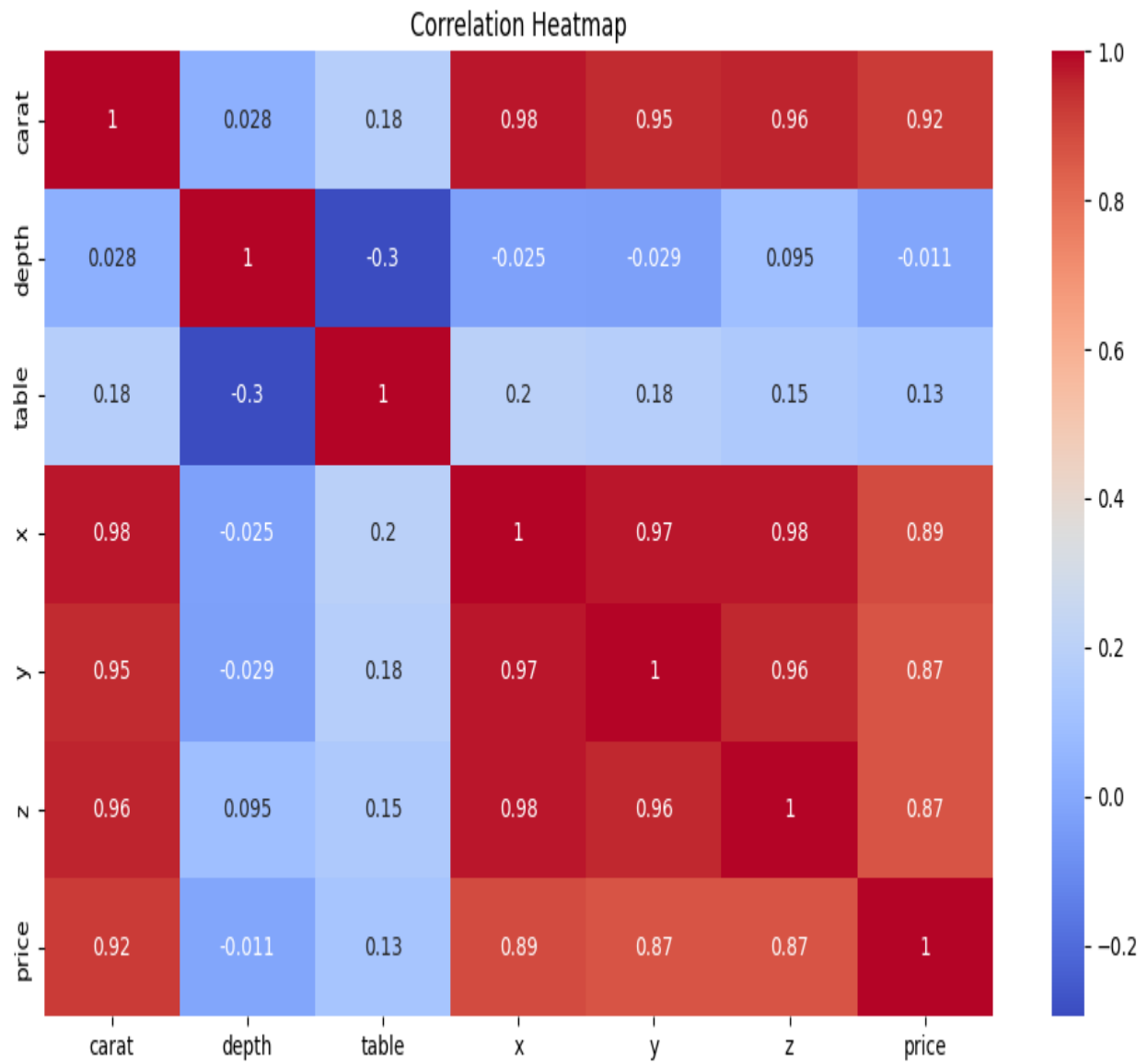


KDE Plot :

The KDE plot for the manual model also forms a bell-like shape, though it may appear slightly wider or asymmetric, reflecting less accurate error distribution. It still confirms that the residuals are generally normal, but the model may have minor deviations from the ideal assumption.



Heatmap:



Source Code : [Click here](#)

Comparison table:

Model	RMSE	R2 Score	Time (s)
Without Scikit-learn	1619.572947	0.836388	0.030868
With Scikit-learn	1460.401776	0.866967	0.030868

Conclusion :

In conclusion, multivariate linear regression is a powerful tool for analyzing the relationship between multiple independent variables and a dependent variable. The main goal of multivariate linear regression is to find the best linear relationship between the independent variables and the dependent variable that minimizes the sum of squared errors. This method can be used to predict the value of the dependent variable based on the values of the independent variables.